

| PLUGINS        |                          | skills / hooks / subagents |
|----------------|--------------------------|----------------------------|
| r-project-core | every R research project | 9 / 13 / 0                 |
| r-package-dev  | R package development    | 3 / 3 / 0                  |
| r-targets      | {targets} pipelines      | 6 / 0 / 1                  |
| r-geospatial   | GIS and spatial analysis | 3 / 0 / 0                  |
| r-spades       | SpaDES modules           | 2 / 0 / 0                  |
| r-landis-ii    | LANDIS-II integration    | 2 / 0 / 0                  |
| r-reporting    | reports and provenance   | 3 / 0 / 0                  |
| r-code-review  | software review          | 1 / 0 / 3                  |

| BUNDLES: RESEARCH, PLUS ONE PER FRAMEWORK |                                                               | 4 |
|-------------------------------------------|---------------------------------------------------------------|---|
| r-bundle-research                         | an R research project                                         |   |
|                                           | project-core, geospatial, reporting, code-review, package-dev |   |
| r-bundle-landis                           | research, plus LANDIS-II                                      |   |
|                                           | research, landis-ii                                           |   |
| r-bundle-spades                           | research, plus SpaDES                                         |   |
|                                           | research, spades                                              |   |
| r-bundle-targets                          | research, plus targets                                        |   |
|                                           | research, targets                                             |   |

| OR ENABLE PLUGINS DIRECTLY |                                |
|----------------------------|--------------------------------|
| R package                  | core, package-dev, code-review |
| {targets} pipeline         | core, targets, code-review     |
| Spatial / GIS              | core, geospatial               |
| produces reports           | add reporting                  |

INSTALL: ADD BOTH MARKETPLACES

1. add the marketplaces

```
/plugin marketplace add FOR-CAST/ai_workflows
```

```
/plugin marketplace add posit-dev/skills
```

Nothing auto-adds another marketplace. r-package-dev depends on r-lib@posit-dev-skills: add both first, in either order, or it and every bundle above it report the dependency missing.

2. enable a bundle, then set policy

```
/plugin install r-bundle-research@ai-workflows
```

```
/r-project-core:project-policy
```

3. share the setup with the project

```
.claude/settings.json
```

 carries `extraKnownMarketplaces` and `enabledPlugins` declare the `posit` marketplace there too, so the dependency resolves. The `claude` plugin CLI does not read project settings: install from a session, once the folder is trusted.

UNINSTALL, OR JUST TURN IT OFF

stop it, remove it, or forget the source

```
/plugin disable r-package-dev@ai-workflows
```

```
/plugin uninstall r-bundle-research@ai-workflows
```

```
/plugin prune
```

```
/plugin marketplace remove ai-workflows
```

`disable` keeps a plugin installed but inert. `uninstall` removes it, and `prune` then drops the dependencies it pulled in that nothing else needs, which is worth running after removing a bundle.

or switch off one piece

One hook: drop its entry from that plugin's `hooks/hooks.json`. All of them: `disableAllHooks` in settings.

One policy rule: remove its key from `r-project-policy.json`.

| THREE KINDS OF TOOLING |                                        |
|------------------------|----------------------------------------|
| skills                 | advice, loaded only when it matches    |
| hooks                  | rules, run on every turn, failing open |
| subagents              | read-only reviewers, on request        |

| SKILLS: WHEN TO REACH FOR THEM |                            | 29 |
|--------------------------------|----------------------------|----|
| <b>r-project-core</b>          |                            |    |
| citation-integrity•            | any DOI or reference       | 9  |
| /commit                        | write the commit           |    |
| design-log                     | work spanning sessions     |    |
| hpc-cluster-runs•              | launch or stop a long run  |    |
| project-config-layout•         | a setting missed a worker  |    |
| /project-policy                | a guard fires wrongly      |    |
| project-tests-ci•              | tests and CI, no package   |    |
| r-style•                       | writing or editing any R   |    |
| root-cause-fixes               | before any fix             |    |
| <b>r-targets</b>               |                            |    |
| targets-cluster-runs•          | tar_make on shared nodes   | 6  |
| targets-debugging              | a target failed or hung    |    |
| targets-project-setup•         | structuring _targets.R     |    |
| targets-spatial•               | terra objects as targets   |    |
| targets-staleness•             | up to date, but is not     |    |
| targets-testing-ci•            | validator, targets CI      |    |
| <b>r-geospatial</b>            |                            |    |
| raster-scale-and-memory•       | raster too big for RAM     | 3  |
| spatial-io-and-crs•            | pointers do not serialise  |    |
| vector-geometry-hygiene•       | empty polygons, slivers    |    |
| <b>r-package-dev</b>           |                            |    |
| package-change-workflow•       | a package you maintain     | 3  |
| package-conventions•           | before air or document()   |    |
| package-dependencies•          | no visible global function |    |
| <b>r-reporting</b>             |                            |    |
| provenance-receipts•           | where data came from       | 3  |
| quarto-reports•                | a render landed oddly      |    |
| report-writing•                | prose someone else reads   |    |
| <b>r-spades</b>                |                            |    |
| module-contract•               | wiring a SpaDES module     | 2  |
| spades-caching                 | re-runs it should not      |    |
| <b>r-landis-ii</b>             |                            |    |
| landis-inputs•                 | before a LANDIS run        | 2  |
| landis-outputs•                | reading a LANDIS map       |    |
| <b>r-code-review</b>           |                            |    |
| verification-method            | before you claim it works  | 1  |

| REVIEW SUBAGENTS              |                        | 4, read-only |
|-------------------------------|------------------------|--------------|
| dependency-auditor            | an undeclared symbol   |              |
| run-forensics                 | a failed or odd run    |              |
| spatial-correctness-reviewer  | CRS and geometry bugs  |              |
| pipeline-invalidation-auditor | silently stale targets |              |

| COMPANIONS, BY WORKFLOW STAGE |                                         |
|-------------------------------|-----------------------------------------|
| plan                          | brainstorming, writing-plans            |
| implement                     | r-package-development, quarto-authoring |
| debug                         | systematic-debugging                    |
| verify                        | verification-before-completion          |
| review                        | receiving-code-review                   |
| release                       | package-change-workflow                 |

Which third-party skills to pair with these, and which to avoid, is in `docs/third-party.md`.

• = paths-scoped: the skill wakes only on matching files. `/name` = slash command only, never auto-invoked. Plugin counts are `skills / hooks / subagents`.

HOOKS: R-PROJECT-CORE14 registrations

A GUARD FIRED, DO THIS INSTEAD

|                                    |                            |   |
|------------------------------------|----------------------------|---|
| SessionStart                       |                            | 1 |
| session-context                    | host, policy, runs, branch |   |
| PreToolUse / Bash                  |                            | 7 |
| advise-bash-hygiene                | a leading cd, a timeout    |   |
| guard-destructive                  | throws away real work      |   |
| guard-host-names                   | a machine name published   |   |
| guard-long-run-interlock           | installs during a live run |   |
| guard-policy                       | installs, air, heavy work  |   |
| guard-process-ownership            | killing what is not yours  |   |
| guard-staging                      | git add -A, commit -a      |   |
| PreToolUse / Edit, Write, Notebook |                            | 3 |
| guard-ascii                        | non-ASCII in R, qmd, bib   |   |
| guard-generated-files              | editing generated files    |   |
| guard-host-names                   | a machine name published   |   |
| PreToolUse / MCP tools             |                            | 1 |
| guard-mcp                          | MCP tools that write       |   |
| PostToolUse / Edit, Write          |                            | 2 |
| check-citations                    | resolves every DOI written |   |
| r-format-and-parse                 | parses and formats your R  |   |

HOOKS: R-PACKAGE-DEV3 registrations

|                           |                            |   |
|---------------------------|----------------------------|---|
| PreToolUse / Edit, Write  |                            | 1 |
| guard-news-dev-heading    | a numbered NEWS heading    |   |
| PostToolUse / Edit, Write |                            | 1 |
| rbuildignore-claude       | .claude into .Rbuildignore |   |
| Stop                      |                            | 1 |
| check-doc-sync            | roxygen ahead of man/      |   |

PROJECT POLICY9 keys

.claude/r-project-policy.json: every rule is off until the repo opts in. Generate one with /r-project-core:project-policy.

|                         |                           |
|-------------------------|---------------------------|
| rBinary                 | note on a bare Rscript    |
| packageInstall          | renv-only installs        |
| noAirFormat             | this repo forbids air     |
| publishRequiresApproval | ask before push, gh, MCP  |
| controllerHosts         | no heavy compute on these |
| hostPatterns            | ask before naming a host  |
| longRunPatterns         | what the interlock sees   |
| heavyCommands           | what counts as heavy      |
| asciiExtensions         | scope of the ASCII guard  |

- ✗ git push --force
  - --force-with-lease
- ✗ git add -A / commit -a
  - stage named paths
- ✗ git reset --hard, clean -f
  - stash, or checkout a named path
- ✗ git branch -D
  - git branch -d
- ✗ install.packages()
  - renv::install("owner/pkg", lock = TRUE)
- ✗ pkill / docker stop
  - check ps, then # owner-verified:
- ✗ heavy job on a controller
  - dispatch it over ssh
- ✗ editing NAMESPACE, man/\*.Rd
  - devtools::document()
- ✗ non-ASCII in .R or .qmd
  - the replacement the hook names
- ✗ installing during a live run
  - wait, or stop the run in order

THE SESSION REPORTSessionStart

- the host, and whether it is a controller
- the project policy that loaded, if any
- the rule: fix the root cause, at file:line
- long runs already going, marked as not yours
- running containers and detached screens
- the branch, and changes that predate the session
- submodule pointers that have drifted

WRITING YOUR OWN HOOK

|          |                                                         |
|----------|---------------------------------------------------------|
| identity | resolve the repo through git, not the working directory |
| deny     | only where a safe alternative exists to take            |
| ask      | for what is fine once a person has looked               |
| context  | to inform without blocking a risky command              |

A deny cannot be approved from the permission prompt; an ask can. If a guard fires when it should not, the fix is almost always a policy value rather than a code change.